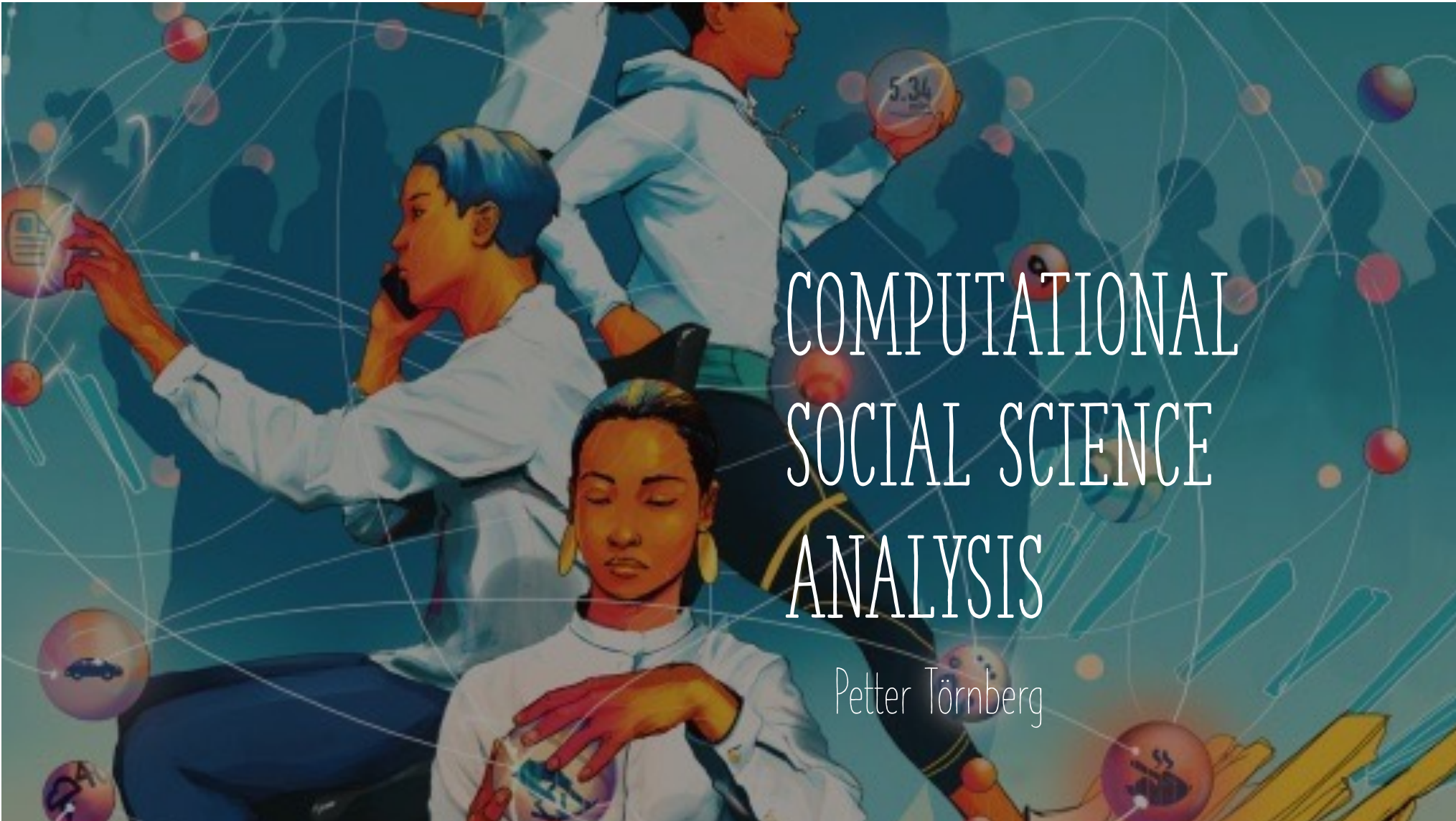




COMPUTATIONAL SOCIAL SCIENCE ANALYSIS

Petter Törnberg

THIS LECTURE

Part 0. Welcome!

Part 1. What is Computational Social Science?

Part 2. How to write a CSS research proposal

PART 0: WELCOME

WHO AM I?

Petter Törnberg (He/him)

Computational Social Science at the Institute for
Logic, Language and Computation.

Background in computer science and physics, and
complexity science.

My research: Social media, polarization and
radicalization, platformization and mediatization.
Large Language Models and AI. Social theory.



WHO IS MICHELLE SCHIMMEL?

A brilliant PhD student in CSS, studying politics through digital data and LLMs.

Your TA and general support person!

Will teach some lectures and workshops.

m.schimmel@uva.nl



WHO ARE YOU?

- Pronouns?
- What did you study?
- Do you know programming?
- What are your research interests?

THIS COURSE

- Practical course to using CSS methods in the social sciences.
- Learn to write a research paper.
- Course mimics the research process:
 1. Research proposal.
 2. Research paper draft.
 3. Peer-review.
 4. Revised final paper
- Each week will have a lecture, a workshop and an independent co-work session.
- Lecture gives theory. Workshop gives code. Independent co-work session gives support and collaboration.
- GitHub: <https://github.com/cssmodels/ComputationalSocialScienceCourse>
- I encourage you to create a course WhatsApp group to discuss!

WHAT YOU CAN EXPECT OF THIS COURSE

- Freedom to pursue your own interest and focus on what you want
- You will learn powerful and useful digital research methods
- A very challenging but exciting course

MY EXPECTATIONS OF YOU

- That you work independently and take your own responsibility for your learning.
- That you start working hard right now
- That you do your own readings and get the resources you need for your project.
- That you ask me for support whenever you need it
- That you submit work on time.
- That you show up: attendance is mandatory (you can miss a maximum of 2/12.)
- That you show up on time! (If you miss the attendance sheet, you missed the class.)

WHAT ARE YOUR EXPECTATIONS?

- Why did you choose to take this course?
- What are you hoping to get out of this course?

COURSE CONTENT

Week 1: What is computational social science? How to write a paper?
+ Learn Programming

Week 2: Scraping and data ethics

Week 3: Natural Language Processing and Large Language Models

Week 4: Social Network Analysis

Week 5: Machine Learning

Week 6: Agent-Based Modeling and Online Experiments

ASSESSMENT

The course is organized around the process of writing a paper.

1. Project proposal - team based (Pass/fail)
2. Research paper draft - team based (Pass/fail)
3. Peer-review on your classmates' papers - individual (20% of grade)
4. Final revised paper - team based (80% of grade)



DUO-GROUP WORK

- The work will now be organized in groups of two.
- It is *your* responsibility to find a collaboration partner!
- Find someone with shared interest and complementary skills.
- Register your group on Canvas.



ASSESSMENT 1: PROJECT PROPOSAL (FROM TODAY; DEADLINE APRIL 13; 11:00)

The primarily goal of this exercise is to make sure that you have an appropriate, feasible, and interesting project idea.

Write a concise one-page document outlining your research project within the realm of Computational Social Science.

What literature are you speaking to? What is your research question? How will you answer it? What are your expectations? What is your data? What is your method?

Write it as you expect to write the final paper, making a guess for what you will find! *Find your story!*

Length: max 1 page (A4, 11pt)

Grading: The grading will be pass/fail. If you fail, you will get the opportunity to revise your plan based on feedback from the instructor.

You will get feedback to guide your project.

See Canvas for more details!

ASSESSMENT 2: RESEARCH PAPER DRAFT (FROM APRIL 13-MAY 20)

The culmination of your project will be a research paper, written in the style of a scholarly article for *Journal of Computational Social Science*.

Present your findings from your project and how they contribute to the literature.

Length: 4000 $\pm$ 200 words in length (excl reference list)

Draft deadline: 2026-05-20 (before noon. 12:00)

See Canvas for more details!

ASSESSMENT 3: PEER REVIEW (FROM MAY 20 - MAY 22. 17:00)

Individual task! You will provide peer-review on another group's papers. Read them carefully and make suggestions on how they can be improved.

Your task is to help the authors get the highest grade possible.

Look at the paper grading criteria and journal guidelines. Imagine that you're me.

Your feedback should be thorough, fair, empathetic, and constructive. The author should feel empowered and supported to improve their paper.

Length: around 1 page in length (A4, Times New Roman, 11pt).

Deadline: May 22, 17:00.

Grading: 1-10. Represents 20% of the final course grade

See Canvas for more details!

ASSESSMENT 4: REVISED RESEARCH PAPER (DEADLINE: MAY 27, 17:00)

- For the final submission, you will revise your paper based on the feedback from the reviewers.
- Final submission should include a 1-page response to each reviewer comment. How did you address the problems they raised? Be kind and grateful for their work helping you with your paper!

Length: 4000 $\pm$ 200 words paper + 500–600 words response to reviewers

Grading: 1–10. Represents 80% of the final course grade

DEADLINE OVERVIEW

Overview assignment deadlines

	Deadline date
Research paper proposal	2026-04-13 (before class begins: 11:00)
Research paper draft (to reviewers)	2026-05-20 (before 12:00)
Peer-reviews	2026-05-22 (before end of workday: 17:00)
Final research paper submission	2026-05-27 (before end of workday; 17:00)
[Paper Resit/Repair]	2026-06-29 (before end of workday; 17:00)

Read the course manual!

Week	Dates	Lecture	Workshop	Project work	Holidays / Notes
1	30 Mar – 3 Apr	What is Computational Social Science? + How to do a research project	Introduction to Python	Programming homework and proposal writing	Good Friday 3 Apr (no teaching Fri)
2	6 Apr – 10 Apr	Acquiring Big Data: Scraping and APIs + Ethics & Law	Web Scraping and APIs in practice	Programming and proposal writing continued	Easter Monday 6 Apr (no teaching Mon)
3	13 Apr – 17 Apr	Natural Language Processing for the Social Sciences	Natural Language Processing in practice	Individual project work	Proposal due Mon 13 Apr, 11:00
4	20 Apr – 24 Apr	Social Network Analysis	Social Network Analysis in practice	Individual project work	
—	27 Apr – 1 May	No teaching or exams			King's Day Mon 27 Apr
5	4 May – 8 May	Machine Learning in the social sciences	Machine Learning in practice	Individual project work	Collective day off Mon 4 May; Liberation Day Tue 5 May
6	11 May – 15 May	Agent-Based Modeling and Online Experiments	Agent-Based Modeling in practice + How to write a research paper	Individual project work	Ascension Day Thu 14 May; Collective day off Fri 15 May
—	18 May – 22 May	<i>No lecture</i>		Draft & peer review	Draft due Wed 20 May 12:00; Peer-reviews due Fri 22 May 17:00
—	25 May – 29 May	<i>Assessment / submission</i>			Whitsun Mon 25 May; Final paper due Wed 27 May 17:00
—	29 Jun				Resit/Repair deadline Mon 29 Jun 17:00

AI POLICY: THIS IS AN AI POSITIVE COURSE!

AI tools are encouraged (e.g., ChatGPT, Claude, GitHub Copilot)

Core principle: understand your work

- You are responsible for all code, analysis, and text
- You must be able to explain and reproduce everything

Transparency requirement

- Add a short note in the Acknowledgements: which tools you used and how

Submission requirements

- Jupyter Notebook with clear structure, comments, and reproducible outputs

Note

- You may be asked to explain your work → impacts your grade
- Misrepresentation = academic misconduct

Acceptable use

Coding, debugging, explaining concepts, improving structure
Generating or refining code that you review and adapt

Unacceptable use

Submitting work you do not understand
Generating entire assignments with AI
Fabricated references or misleading results

IT'S YOUR COURSE!

I will update the course as we go along.

Do you want to cover something in particular?

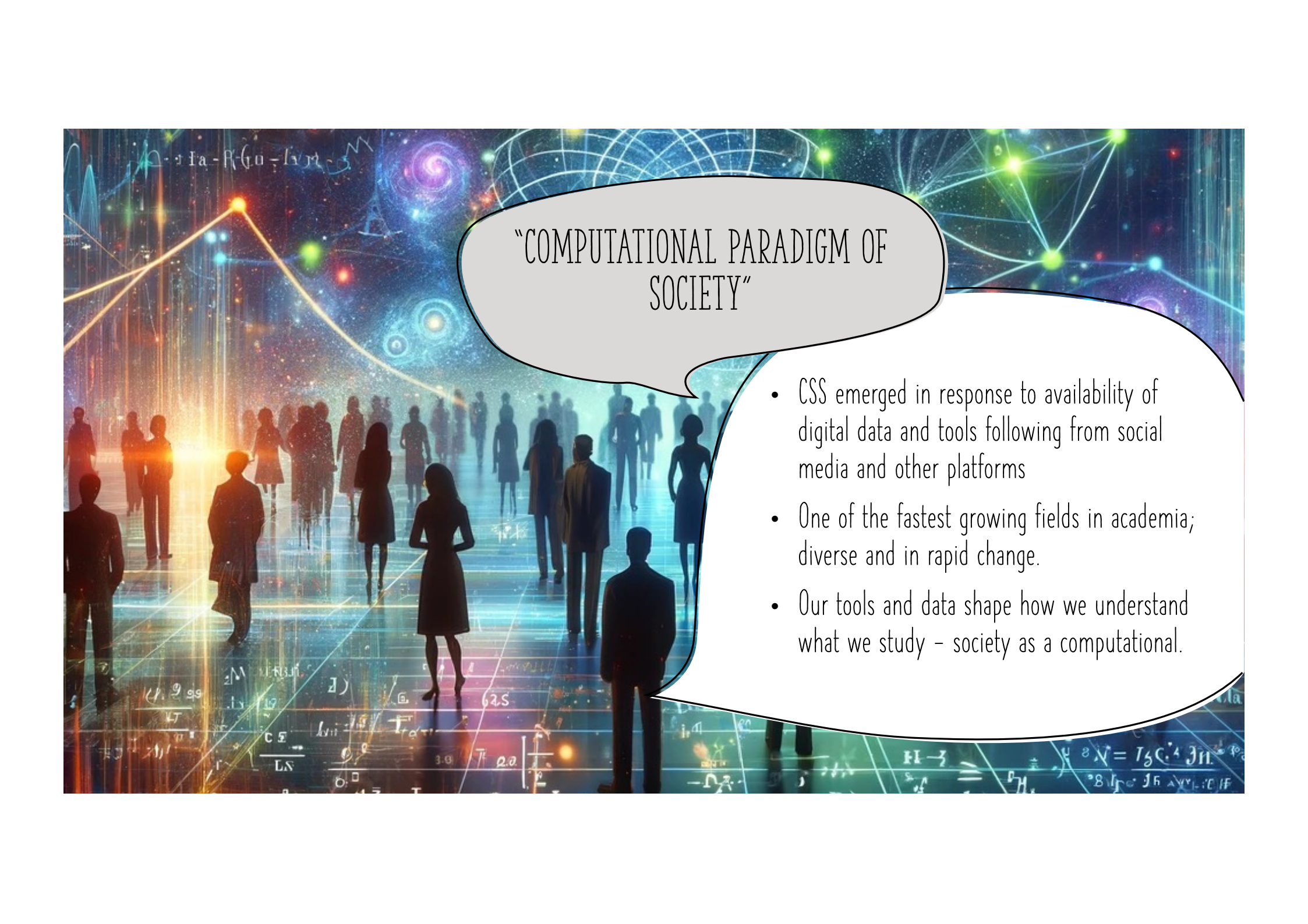
For example: you want to move the deadlines? You want to change the order of the lectures? Want to learn more about something? You want me to talk less?

We can do it. Just let me know!



PART 1: WHAT IS
COMPUTATIONAL SOCIAL
SCIENCE?





"COMPUTATIONAL PARADIGM OF SOCIETY"

- CSS emerged in response to availability of digital data and tools following from social media and other platforms
- One of the fastest growing fields in academia; diverse and in rapid change.
- Our tools and data shape how we understand what we study – society as a computational.

'Digital is what gave culture the scale of physics, chemistry or neuroscience. Now we have enough data and fast enough computers to actually study the "physics" of culture'
Lev Manovich 2016





"just as the invention of the telescope revolutionized the study of the heavens, so too by rendering the unmeasurable measurable, the technological revolution in mobile, Web, and Internet communications has the potential to revolutionize our understanding of ourselves ... we have finally found our telescope. Let the revolution begin"

Duncan Watts 2011: 266

WHAT TYPE OF DATA DOES CSS USE?

Data is not neutral: data + methods shape epistemology

Each data type implies a different way of seeing and knowing the social world

Types of data that have shaped approaches to social science:

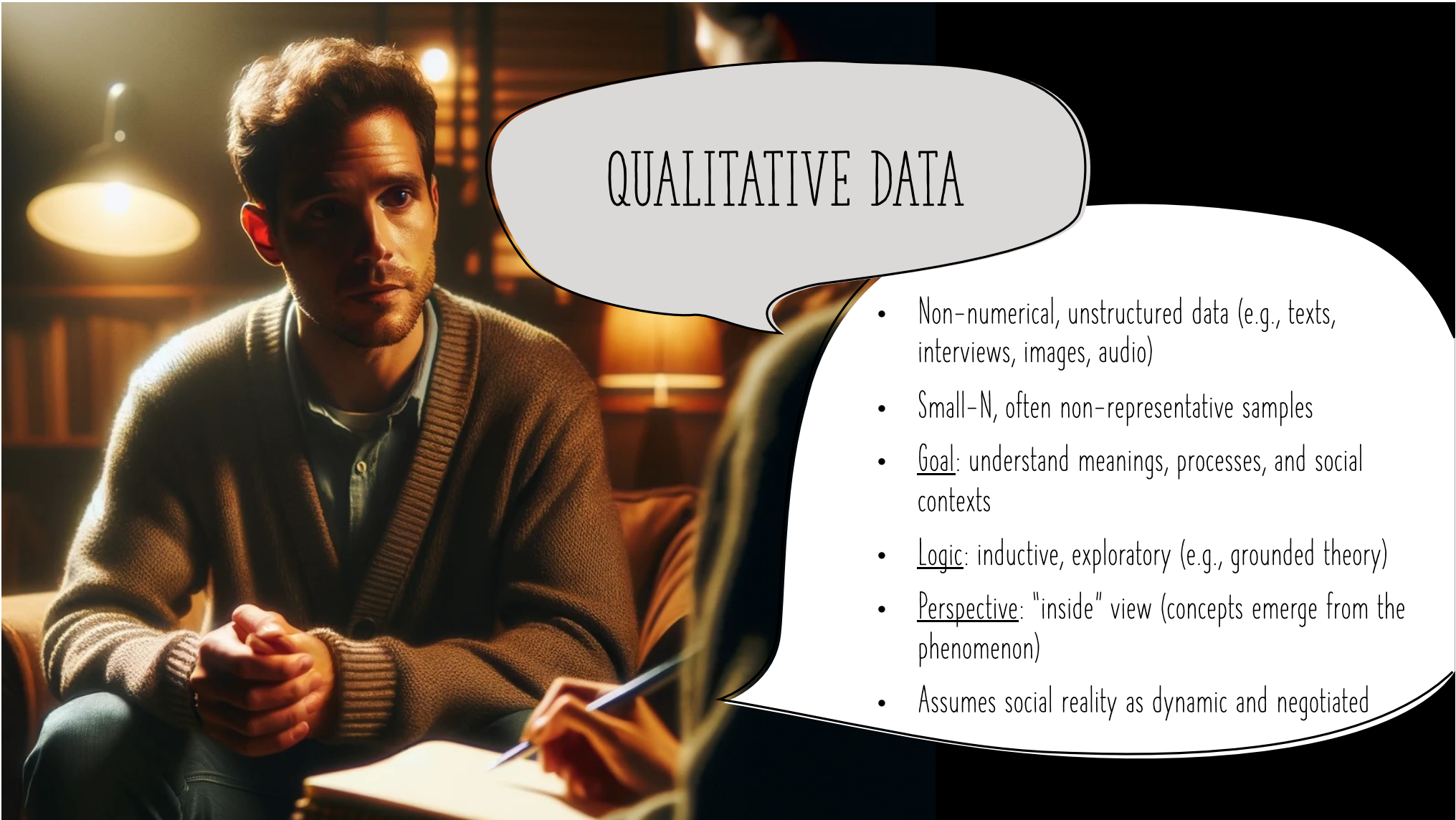
- Quantitative
- Qualitative
- Digital

QUANTITATIVE DATA

	Variables				
	Age	Sex	Income	...	etc.
Cases					

Figure 4.1 A data matrix for variable analysis

- Typically collected through surveys of (ideally) representative samples
- Structured data: rows = individuals, columns = variables (numeric or categorical)
- Goal: identify relationships between variables and generalize to populations
- Logic: deductive, confirmatory
- Perspective: "outside" view (observer-driven categories)



QUALITATIVE DATA

- Non-numerical, unstructured data (e.g., texts, interviews, images, audio)
- Small-N, often non-representative samples
- Goal: understand meanings, processes, and social contexts
- Logic: inductive, exploratory (e.g., grounded theory)
- Perspective: "inside" view (concepts emerge from the phenomenon)
- Assumes social reality as dynamic and negotiated



DIGITAL DATA

- Generated through platforms and digital interactions
- Includes digital trace data (e.g., clicks, posts, networks, logs).
- Not representative sample, not structured.
- Often large-scale, behavioral, and continuously generated
- Fits neither in qual or quant. Requires new approaches: computational social science, NLP, network analysis
- Blurs boundaries: combines scale (quantitative) with richness (qualitative)
- Raises new challenges: bias, platform effects, data access

Another Perspective of TYPES of DATA

1. Attribute data:

Eg., survey, demographics

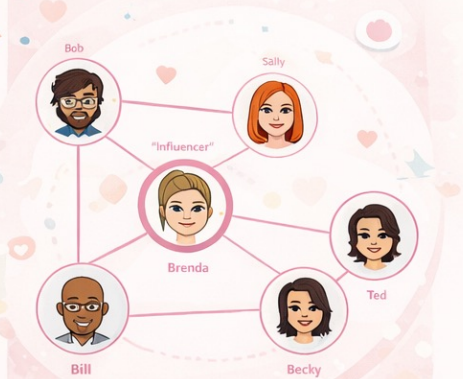
	Variables			
	Age	Sex	Income	... etc.
1	25	F	\$50k	
2	3	M	\$530k	
3	4	J	\$Tak	
4	5			
...	etc.			

2. Ideational data:

E.g., text, speech,
images, videos..

3. Relational data:

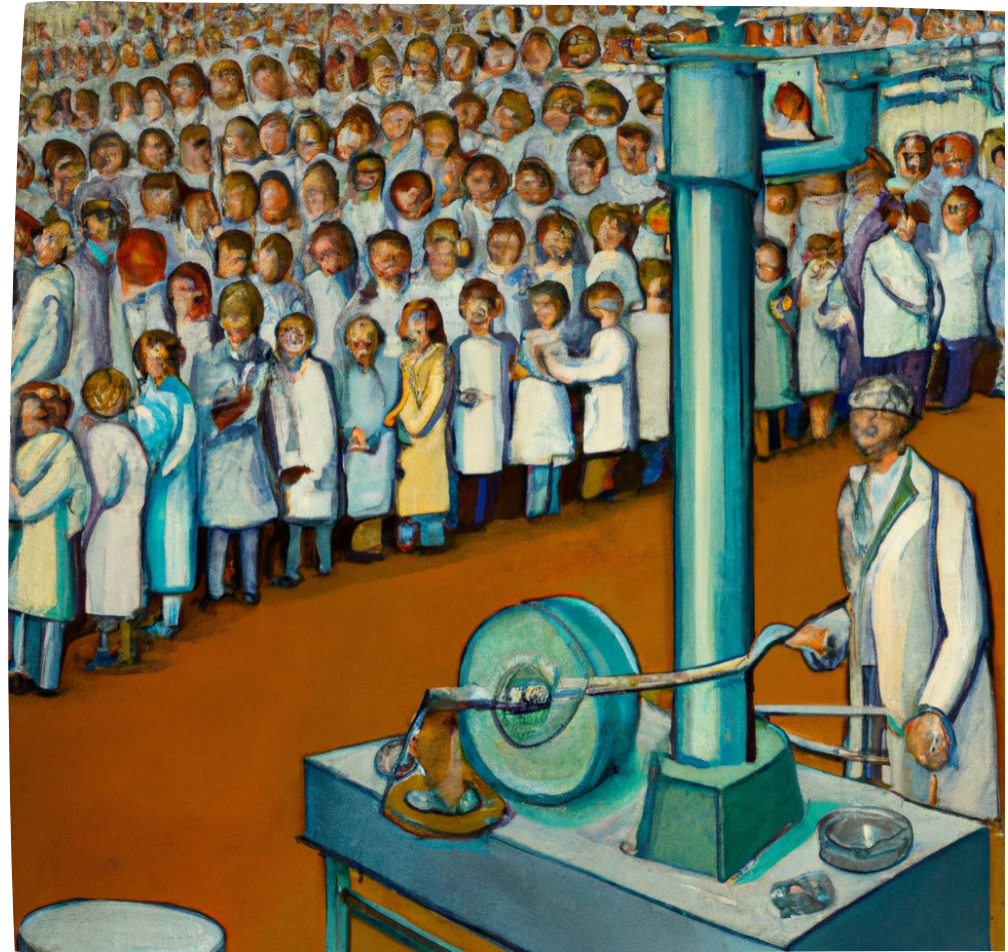
E.g., friendships,
connections, etc.



BIG DATA ANSWERS SOME LONG-STANDING CRITIQUE OF QUANTITATIVE PERSPECTIVE

"For the last thirty years, empirical social research has been dominated by the sample survey. But as usually practiced, ... the survey is a sociological meat grinder, tearing the individual from his social context and guaranteeing that nobody in the study interacts with anyone else in it."

Allen Barton, 1968





QUANTITATIVE DATA IS MADE FOR STATISTICS

- Quantitative methods squeeze the world into mold that fits the method.
- Variables are taken to represent some aspect of the world that the researcher believes is relevant for a phenomenon.
- *Designed* for statistical analysis: variable-variance analysis. "Gender". "Employment."

The background of the slide is a digital landscape. It features rolling hills and mountains in shades of blue, brown, and gold. The foreground is filled with a dense field of binary code (0s and 1s) in various colors (blue, orange, green, white). The overall aesthetic is futuristic and data-driven.

IS BIG DATA 'UNFILTERED' OR 'RAW'?

Big data shows a different type of world: relational, dynamic.

Some researcher suggest that Big Data are "naturally occurring by-product" of social processes, rather than something produced for scientific consumption (Lazer et al., 2020). "Footprints" or "traces".

But there is no such thing as raw data. Big data are made for algorithmic processing (Marres, 2017).

Big data is not necessarily bigger, but *different* in structure and format: fits poorly into both quant/qual paradigms.



“DATA CAPITALISM” OR “SURVEILLANCE CAPITALISM”

- Data are also a form of commodity; “the new oil”.
- “Datafication”: processes to extract information about individuals to market to data capitalist companies.
- “Dataism”: the ideology that gives data its value.
- Computational Social Science can be understood as part of this mode of capitalism.

Törnberg, P., & Uitermark, J. (2021). For a heterodox computational social science. *Big Data & Society*, 8(2).

We need to be aware of the context in which the data were produced.

Platforms are not neutral; they shape the structure of social life, and we are in part measuring the effects of these platforms.

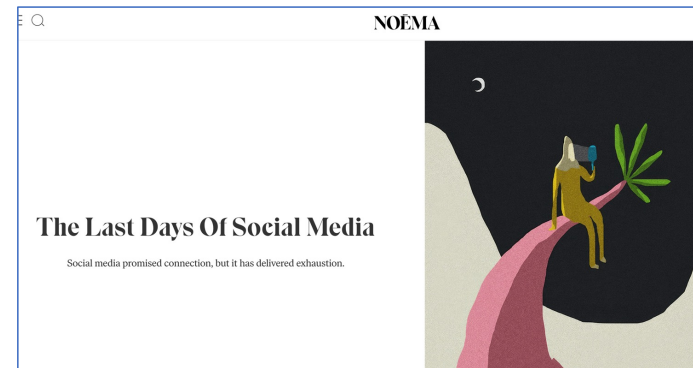


DISCUSSION: IS SOCIAL MEDIA ENDING?

CSS emerged in response to social media, with behavioral people talking online as a data source.

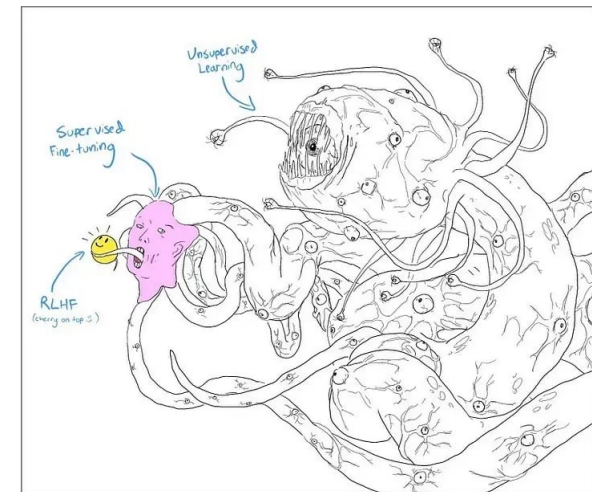
But now many argue that the social media is coming to its end: LLMs are replacing users. We're watching TikTok videos instead of talking on X/Twitter.

What do you think will happen to CSS? Does the discipline still make sense?



LARGE LANGUAGE MODELS ARE TRANSFORMING CSS

- A new research tool
 - Versatile for text analysis, coding, classification, and data generation
 - Extremely powerful - and increasingly easy to use
 - Will be a core part of this course
- A new object of study
 - LLMs are reshaping how people seek information and interact with knowledge
 - Human-AI interaction is becoming a major domain of social behavior
- A shift in digital behavior
 - More people now interact with LLMs (>50%) than actively post on social media (~30%)
 - Signals a move from networked communication → AI-mediated interaction



AN UNCERTAIN FUTURE: FROM CSS TO AI?

Shifting identities

- Growing number of scholars rebranding from *Computational Social Science* → *AI / LLM research*
- Auditing LLMs and AI: bias
- LLMs as a new form of media
- Signals a broader transformation of the field

Field in flux

- Is this a continuation of CSS – or a paradigm shift?
- What remains distinctly "social science"?



CSS FOR THE PURPOSES OF THIS COURSE

- We view CSS as a flexible label. Make it your own!
- CSS allows us to explore the boundaries outside the conventional methods: interpretation and exploration in large-scale data.
- "Heterodox CSS"
- You can combine CSS methods with a quant or a qual approach – or go beyond both!
- Use AI as method or as object of study!

what are other
words for
heterodox?



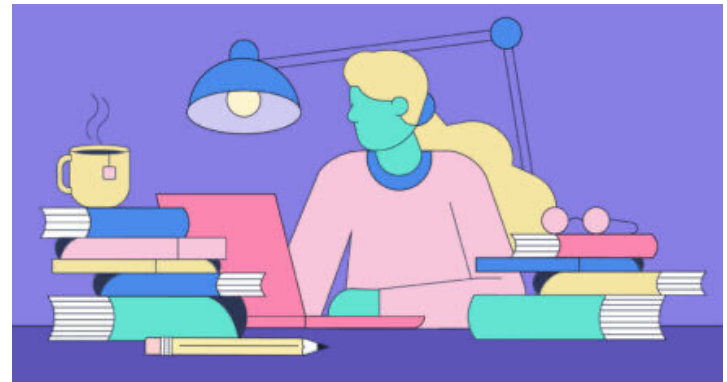
heretical, unorthodox,
dissident, iconoclastic,
nonconformist, unconventional,
dissenting, apostate



IN SUMMARY: COMPUTATIONAL SOCIAL SCIENCE

- CSS is a new interdisciplinary field, in between computer science, social science and statistics.
- Characterized by use of digital data and computational methods to study social world
- Closely linked to big tech and Silicon Valley
- Neither conventional quantitative or qualitative, but shaped by the distinct epistemology of the digital: interactional, dynamic, fluid, relational, clusters...
- It's still being defined. You can be part of shaping it!

PART 2: HOW TO WRITE A CSS RESEARCH PROPOSAL



WHAT ARE APPROPRIATE TOPICS?

- Anything goes! Pick something you are interested in (and knowledgeable about.)
- Doesn't have to be a "computational" or "digital" topic.
- Are you part of some specific niche or community that is not broadly known? Might make for a good topic.
- Be unique and creative!

Be creative!

WHAT DATA CAN YOU USE?

We're looking for texts, relationships, or other forms of digital data.

Advice: use existing data, scraping new data will take too long!

– Kaggle is a useful source of data.

I can offer: Historical political manifestos. Social media posts from parties across the world. Stormfront.org

- All of Reddit is easy to download - but hard to digest. TikTok, BlueSky and YouTube are relatively easy to scrape.
- Airbnb is free and easy: Insideairbnb.com
- Parliamentary speeches
- Newspapers, headlines.
- Global Event database (GDELT), Global terrorism database
- Online experiments (e.g., MTurk)
- ... what else?

Be creative!

WHAT METHODS CAN YOU USE?

What we cover in this course:

- Natural Language Processing (Vectorization, Sentiment analysis, Topic modeling...)
- Large Language Models
- Social Network Analysis
- Machine Learning
- Online Experiments
- Agent-Based Modeling

WHAT CHARACTERIZES A GOOD RESEARCH QUESTION?

1. **Clarity and specificity:** It should be formulated in a way that leaves no ambiguity about what you are investigating.
2. **Feasibility:** Possible to answer the research question within the limited time and resources
3. **Relevance:** Should contribute to the existing body of knowledge or address a gap in the research.
4. **Researchability:** Should actually be possible to answer.
5. **Originality:** Should address a gap in the literature, propose a new idea, or explore understudied area.
6. **Pragmatism:** You need to be able to write a paper regardless of what you find.

TYPES OF RESEARCH QUESTIONS

1. Descriptive: These questions aim to describe the characteristics of a phenomenon..

Example: What is the level of toxicity of political conversations on Twitter, BlueSky and Mastodon?

2. Causal (Explanatory) : Identify a cause-and-effect relationship.

Example: Did Elon Musk's takeover of Twitter make the platform more toxic?

3. Exploratory: Open-ended questions aiming to learn about a little-known phenomenon.

Example: What linguistic factors of a Twitter message influence how much engagement it receives?

EXAMPLES OF BAD RESEARCH QUESTIONS

1. "How do media affect society?" [Vague and too broad in scope; not feasible]
2. "How does changing the Facebook algorithm affect toxicity?" [Not feasible for the course]
3. "Which city has the best pizza?" [Limited academic value]
4. "Can sentiment analysis of Twitter data predict stock market trends?"
[Fine, but hard to prove a negative. What do you write about if you fail?]

EXAMPLE IDEAS

- "Is Cunningham's Law true? An online experiment on Reddit"
- "How is 'AI' emerging as a political issue across party families in their manifestos in Europe?"
- "Have mainstream news headlines have become more 'clickbaity' over the last 20 years?"
- "How is the Israel-Palestine conflict framed differently in newspapers across countries?"

Other ideas?

DO'S AND DON'TS OF WRITING A RESEARCH PAPER

Do tell a story

- Writing a paper is storytelling!
- Needs to be something surprising; something interesting.
- Your job is also to be a bit of a journalist. Tell a story!
- "Hourglass" structure: Start broadly, narrow down to specific research questions or hypotheses, then broaden out again to discuss the implications of your findings.

Schimmel. "Writing Science: How to Write Papers That Get Cited and Proposals That Get Funded"



Do use theory

- Use theory to *tell your story*.
- Theory is our tool, and our language
- Focus on using theory to engage with a topical and broadly interesting discussion.

Do work with middle-range theory

- We contribute not to grand theories, nor on the smallest theory, but on the middle-range
- Don't: "I'm going to test whether Goffman's theory of self-presentation is true"
- Do: "How do content recommendation algorithms on YouTube contribute to the formation of echo chambers?"

Do look at examples of similar papers

- Look at other papers doing something similar
- See how they structure their story
- Draw inspiration! (But don't plagiarize)

Do speak to your strengths

- The aim here is to produce a paper that tells a good story – so don't do something that will be unnecessarily hard and unrewarding.
- Draw on your strengths and what you know. For instance, you are younger than 99% of researchers, which gives you unique insights!
- What's some community or group that you have insider insights into?
- Do something interesting with limited time and resources!

Don't be too ambitious

- Papers are often really small: they nail down a small but broadly important claim
- Most common problem with student work is that it does too much!.



FOR REFERENCE: ARCHETYPICAL STORIES

1. **Gap Identification Structure:** "While significant progress has been made in understanding X, a gap remains in our knowledge of Y. In this study, we address this gap by investigating Z."
2. **Problem-Solution Structure:** "The major problem in field X is Y. This paper proposes a novel solution, Z, which overcomes limitations of existing approaches by..."
3. **Contradiction Resolution Structure:** "The dominant idea in the field is X, but recent evidence suggests Y, creating a contradiction. Our research resolves this contradiction by demonstrating that Z."
4. **Sequential Development Structure:** "To date, research in X has progressed from understanding basic elements (A) to more complex components (B). Building on this knowledge, we introduce a new dimension, C, which advances the field by..."
5. **Comparative Analysis Structure:** "Several theories exist about phenomenon X—Theory A suggests Y, while Theory B suggests Z. Through our comparative analysis, we identify the strengths and weaknesses of each and propose a unified theory that..."
6. **Evolutionary Structure:** "Historically, the understanding of X has evolved from A to B. We extend this evolution by introducing C, a concept that not only builds on previous work but also opens new research avenues by..."
7. **Question-Driven Structure:** "A key question in the study of X is 'how does Y affect Z?' This paper addresses this question through a series of experiments/analyses, revealing that..."
8. **Debate Engagement Structure:** "The field of X is characterized by a longstanding debate between proponents of Y and advocates of Z. We engage with this debate by providing new evidence that suggests..."
9. **Innovation Introduction Structure:** "Current methodologies for studying X are limited by Y. To overcome these limitations, we introduce an innovative approach, Z, which allows for more accurate/efficient investigation by..."
10. **Theory Extension Structure:** "Theory X provides a comprehensive framework for understanding Y. However, it falls short in explaining Z. Our work extends Theory X by incorporating W, thereby offering a more complete understanding of..."

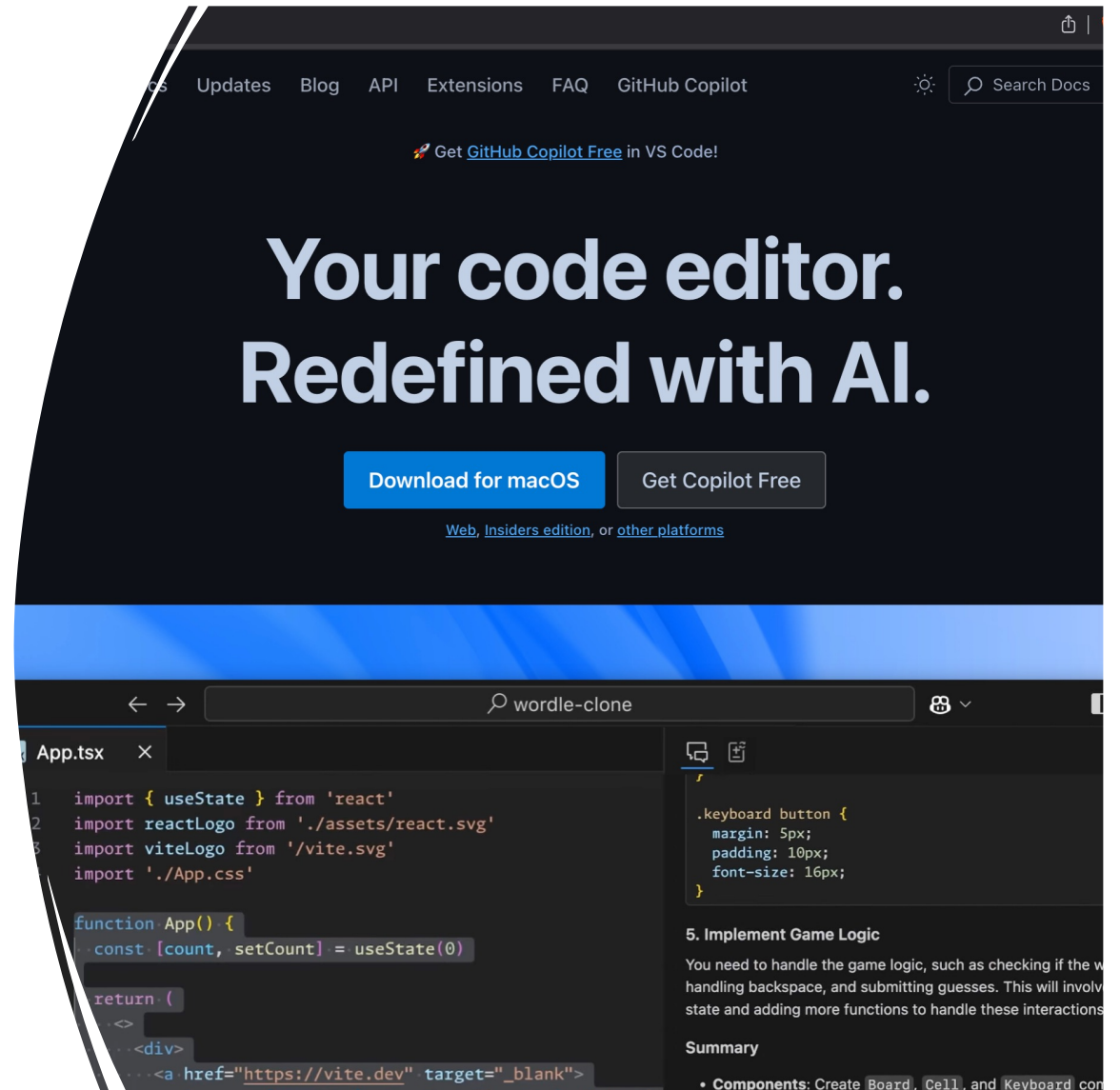


HOMEWORK

First two weeks will be proposal writing + programming.

Homework 1 until Wednesday

1. Install Visual Studio Code
2. Get it running with Jupyter Notebook + Python
3. Find out which is the most common word in Marx's *Capital*.



Homework 2: Read the 3 example papers from last year

For inspiration, Github has three papers from students from last year - to inspire and illustrate what you can do!

But don't limit yourself!

Homework 3

- Think up a project idea!
- Find a teammate
- Register your team on Canvas

Friday independent work session

- You have space on Thursdays 13:00–17:00 to work in the JK building (see JK3.02)
 - Collaborate, help each other, or just work in an inspiring environment.
 - Michelle Schimmel will be there to help
-
- Excellent opportunity to discuss project ideas and find a teammate!